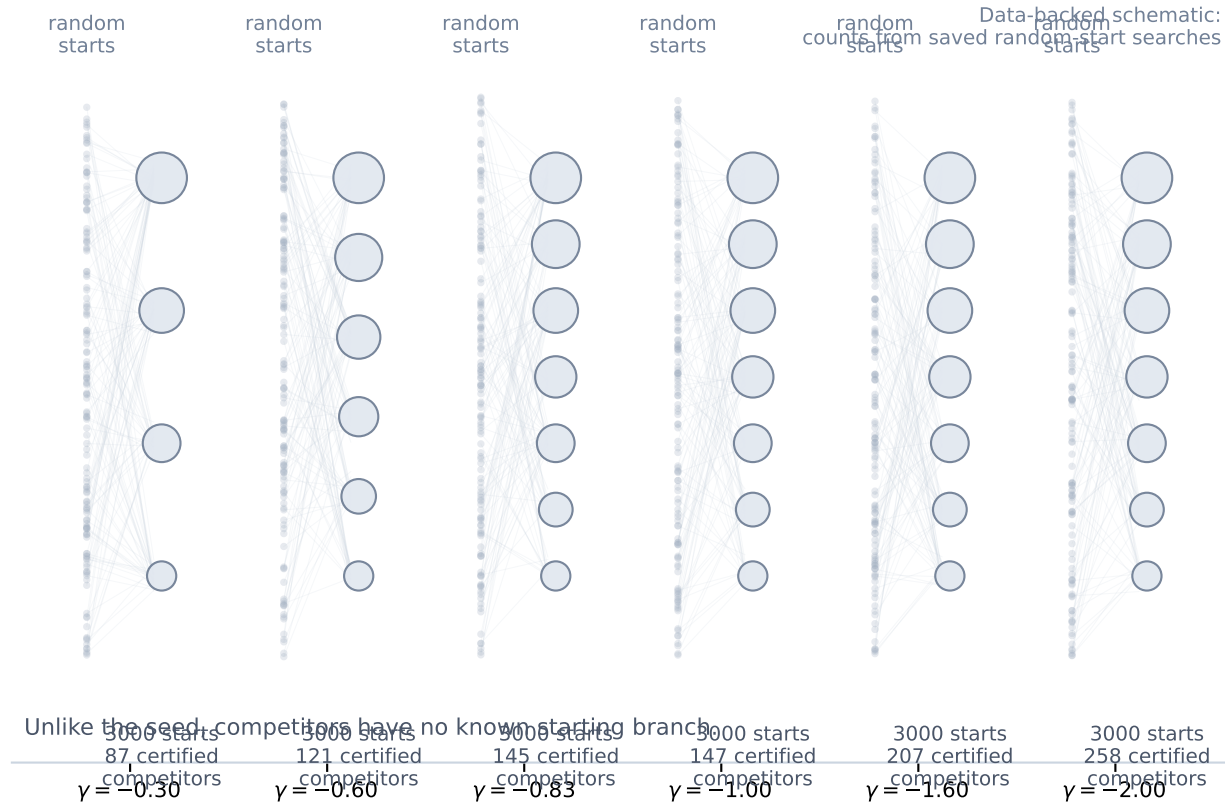


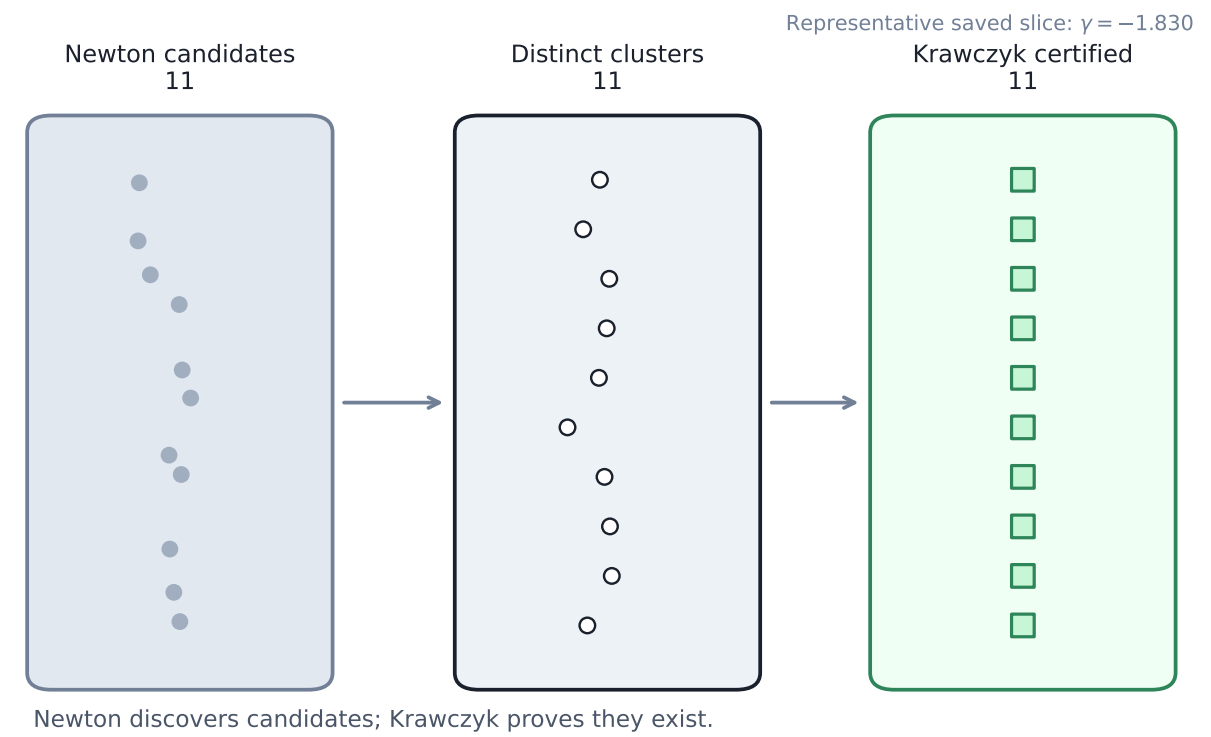
From competitor search to certified conjugate saddle pair

1. Explore the saddle landscape

Many Newton starts are needed because competitors have no known starting branch

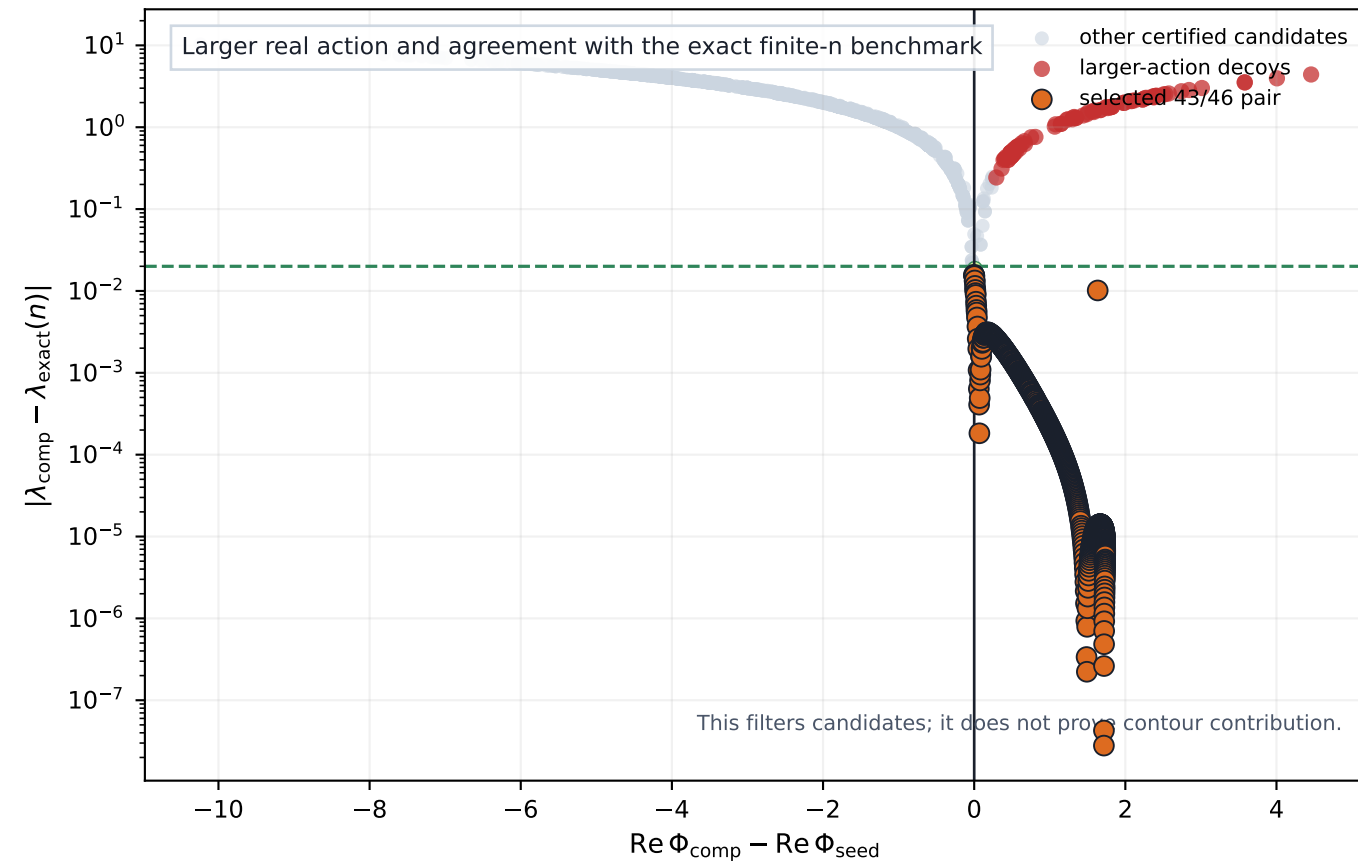


2. Validate candidate roots



3. Identify relevant competitors

Larger real action is not enough, relevance also needs exact-benchmark agreement



4. Continue the selected branch

Use each certified root to initialize and certify the next γ -value

